King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Major Exam I Term 221

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. Consider the four points $A(0,3,-4)$, $B(2,2,-3)$, $C(0,6,-6)$ and $D(-2,10,a)$. Suppose these four points lie in the same plane. Then $a =$

(a) -10

(b) -9

(c) 3

(d) -8

(e) 7

2. The exact length of the polar curve

$$r = 2\theta^2, \quad \theta \in [0, \pi]$$

is

(a) 8

(b) 16

(c) $\frac{2}{3}\left(\pi^3 - 8\right)$

(d) $\frac{2}{3}\left((\pi^2+4)^{3/2} - 8\right)$

(e) $(\pi^2+4)^{3/2}$

3. The surface area of the solid generated by rotating the curve given by the parametric equations

$$x = 6t, \quad y = 3t^2, \quad 0 \le t \le 1,$$

about the y-axis is

(a) $3\pi\left[2 - \sqrt{2}\right]$

(b) $24\pi\left[2\sqrt{2} - 1\right]$

(c) $12\pi\left[\sqrt{2} - 1\right]$

(d) $6\pi\left[\sqrt{2} + 1\right]$

(e) $9\pi\left[1 - 2\sqrt{2}\right]$

4. The area of the region enclosed by the polar curve $r = 2 - \cos\theta$ is

(a) $\dfrac{11\pi}{2}$

(b) $\dfrac{7\pi}{2}$

(c) $\dfrac{5\pi}{2}$

(d) $\dfrac{9\pi}{2}$

(e) $\dfrac{3\pi}{2}$

5. The polar equation $r = \tan\theta \sec\theta$ represents a

(a) Line

(b) Ellipse

(c) Hyperbola

(d) Parabola

(e) Circle

6. If the angle between the vectors $\langle 2, 1, -1\rangle$ and $\langle 1, x, 0\rangle$ is $\dfrac{\pi}{4}$, then $x =$

(a) $1 \pm \sqrt{6}/2$

(b) $1 \pm \sqrt{3}$

(c) $-1 \pm \sqrt{3}/2$

(d) $2 \pm \sqrt{6}$

(e) $1 \pm \sqrt{6}$

7. The graph of the polar curve $r = 3\cos\theta$ has a vertical tangent line when $(r, \theta) =$

(a) $(0, \pi/2)$

(b) $(3, \pi/2)$

(c) $(0, 2\pi)$

(d) $(0, \pi/4)$

(e) $(0, \pi)$

8. If $\vec{a} = \langle 3, 0, -1 \rangle$, then a vector $\vec{b} = \langle 0, u, v \rangle$ such that $\text{comp}_{\vec{a}}\, \vec{b} = 2$ is

(a) $\langle 0, 10, -2\sqrt{10} \rangle$

(b) $\langle 0, -1, 2\sqrt{10} \rangle$

(c) $\langle 10, 10, 3\sqrt{10} \rangle$

(d) $\langle 0, 0, 3\sqrt{10} \rangle$

(e) $\langle 0, 1, -\sqrt{10} \rangle$

9. The unit vectors that are parallel to the tangent line to the curve $y = 2x^2 - x - 1$ at the point $(1, 0)$ are

(a) $\langle 1, 0 \rangle$ and $\langle -1, 0 \rangle$

(b) $\langle 0, 1 \rangle$ and $\langle 0, -1 \rangle$

(c) $\left\langle \frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}} \right\rangle$ and $\left\langle \frac{-3}{\sqrt{10}}, \frac{-1}{\sqrt{10}} \right\rangle$

(d) $\left\langle \frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}} \right\rangle$ and $\left\langle \frac{-1}{\sqrt{10}}, \frac{-3}{\sqrt{10}} \right\rangle$

(e) $\left\langle \frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$ and $\left\langle \frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}} \right\rangle$

10. The curve given parametrically by

$$x = \tan\theta, \quad y = 2\sec\theta, \quad -\pi/2 < \theta < \pi/2$$

is

(a) a parabola with vertical axis of symmetry

(b) a straight line

(c) a hyperbola

(d) an ellipse

(e) a parabola with horizontal axis of symmetry

11. The area of the region that lies inside the curve $r = 1 - \sin\theta$ and outside the curve $r = 1$ is

(a) $\pi - 2$

(b) $\pi + 2$

(c) $\pi + \frac{1}{2}$

(d) $\frac{\pi}{2} + 2$

(e) $\frac{\pi}{4} + 2$

12. The area of the region enclosed by the curve that is given by the parametric equations $x = 2\cos\theta,\ y = \frac{1}{2}\sin\theta,\ 0 \le \theta \le 2\pi$, is

(a) 0

(b) 3π

(c) 2π

(d) π

(e) 4π

13. Consider for $a \ge 0$ the following three points

$$A(-2,0,0), \quad B(-1,1,3), \quad C(-2,a,2).$$

Assume that the area of the triangle $\triangle ABC = \sqrt{6}$. Then $a =$

(a) $\frac{3}{5}$

(b) 0

(c) 2

(d) $\frac{1}{5}$

(e) 1

14. A particle moves three times, counterclockwise, along the circle

$$x^2 + (y-1)^2 = 9.$$

If the particle starts at the point $(3,1)$, then the parametric equations of the path of the particle are

(a) $x = 3\cos t,\ y = 1 - 3\sin t,\ 0 \le t \le 6\pi$

(b) $x = 3\cos t,\ y = 1 + 3\sin t,\ 0 \le t \le 6\pi$

(c) $x = 3\cos(3t),\ y = 1 - 3\sin(3t),\ 0 \le t \le 2\pi$

(d) $x = 3\sin(3t),\ y = 1 + 3\cos(3t),\ 0 \le t \le 2\pi$

(e) $x = 3\sin t,\ y = 1 + 3\cos t,\ 0 \le t \le 6\pi$

15. The slope of tangent line to the polar curve $r = \frac{1}{\theta}$ at $\theta = \pi$ is given by

(a) 2

(b) 0

(c) $-\pi$

(d) -2

(e) π

16. Suppose that two vectors $\vec{a}, \vec{b}$ satisfy

$$\vec{a} \cdot \vec{b} = \sqrt{15}, \qquad \vec{a} \times \vec{b} = \langle -2, 0, -1 \rangle.$$

Then the angle between $\vec{a}$ and $\vec{b}$ is

(a) $\dfrac{2\pi}{5}$

(b) $\dfrac{\pi}{2}$

(c) $\dfrac{\pi}{4}$

(d) $\dfrac{\pi}{3}$

(e) $\dfrac{\pi}{6}$

17. The vector component of $\vec{v} = \langle 1, 1, 1 \rangle$ orthogonal to $\vec{b} = \langle 2, 2, 0 \rangle$ is

(a) $-\hat{j}$

(b) $\hat{k}$

(c) $-\hat{i}$

(d) $\hat{j}$

(e) $\hat{i}$

18. The curve given by the parametric equations

$$x = t^3 - 3t \quad \text{and} \quad y = t^2$$

is

(a) concave up when $t > 1$

(b) concave down when $-\dfrac{1}{2} < t < \dfrac{1}{2}$

(c) concave up when $t > 0$

(d) concave down when $t < 0$

(e) concave up when $-1 < t < 1$

19. If M is the distance from the point $P(3, -2, 4)$ to the xz-plane and m is the distance from $P(3, -2, 4)$ to the y-axis, then $M + m$ is equal to

(a) 7

(b) 6

(c) 3

(d) 8

(e) 4

20. The curve given by the parametric equations

$$x = 2\cos^3\theta, \qquad y = 2\sin^3\theta, \quad 0 \le \theta < \frac{7\pi}{4}$$

has

(a) only one horizontal and only one vertical tangent lines

(b) only one horizontal tangent line and no vertical tangent line

(c) two horizontal and one vertical tangent lines

(d) no horizontal tangent line and only one vertical tangent line

(e) one horizontal and two vertical tangent lines

Answer Key

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Q	Ans	Brief explanation
1	**(b)**	Triple product $= 6a + 54 = 0$, so $a = -9$.
2	**(d)**	$L = \int_0^{\pi} 2\theta\sqrt{\theta^2+4}\,d\theta = \frac{2}{3}\left((\pi^2+4)^{3/2} - 8\right)$.
3	**(b)**	$ds = 6\sqrt{1+t^2}dt$: $S = 36\pi\int_0^1 2t\sqrt{1+t^2}dt = 24\pi(2\sqrt{2}-1)$.
4	**(d)**	$A = \int_0^{\pi}(2-\cos\theta)^2 d\theta = \frac{9\pi}{2}$.
5	**(d)**	$r\cos^2\theta = \sin\theta \Rightarrow x^2 = y$: a parabola.
6	**(a)**	$2 + x = \sqrt{3}\sqrt{1+x^2} \Rightarrow 2x^2 - 4x - 1 = 0$, $x = 1 \pm \frac{\sqrt{6}}{2}$.
7	**(a)**	$x = 3\cos^2\theta$, $x' = -3\sin 2\theta = 0$: vertical tangent at $(r,\theta) = (0, \pi/2)$.
8	**(a)**	comp $= \frac{-v}{\sqrt{10}} = 2 \Rightarrow v = -2\sqrt{10}$ (u free): $\langle 0, 10, -2\sqrt{10}\rangle$.
9	**(d)**	Slope $y'(1) = 3$, direction $\langle 1, 3\rangle$: $\pm\frac{1}{\sqrt{10}}\langle 1, 3\rangle$.
10	**(c)**	$1 + x^2 = y^2/4$, i.e. $\frac{y^2}{4} - x^2 = 1$: a hyperbola.
11	**(e)**	Outside $r = 1$ for $\pi < \theta < 2\pi$: $\frac{1}{2}\int_{\pi}^{2\pi}(\sin^2\theta - 2\sin\theta)d\theta = \frac{\pi}{4} + 2$.
12	**(d)**	Ellipse $\frac{x^2}{4} + 4y^2 = 1$: $A = 4\int_0^{\pi/2}\sin^2\theta\,d\theta = \pi$.
13	**(c)**	$\overrightarrow{AB} \times \overrightarrow{AC} = \langle 2-3a, -2, a\rangle$; $10a^2 - 12a + 8 = 24 \Rightarrow a = 2$.
14	**(b)**	Centre $(0,1)$, $r = 3$: $x = 3\cos t$, $y = 1 + 3\sin t$, $0 \le t \le 6\pi$ (three turns).
15	**(c)**	$\frac{dy}{dx} = \frac{-\sin\theta + \theta\cos\theta}{-\cos\theta - \theta\sin\theta}$ at $\theta = \pi$ equals $-\pi$.
16	**(e)**	$\tan\theta = \frac{\|a\times b\|}{a\cdot b} = \frac{\sqrt{5}}{\sqrt{15}} = \frac{1}{\sqrt{3}}$, so $\theta = \frac{\pi}{6}$.
17	**(b)**	proj $= \langle 1, 1, 0\rangle$, so $\vec{v} - \text{proj} = \langle 0, 0, 1\rangle = \hat{k}$.
18	**(e)**	$y'' = \frac{-2(t^2+1)}{9(t^2-1)^3} > 0$ exactly for $-1 < t < 1$: concave up there.
19	**(a)**	$M = \|y\| = 2$, $m = \sqrt{x^2+z^2} = 5$, so $M + m = 7$.
20	**(a)**	$dy/dx = -\tan\theta$: slope 0 gives only $y = 0$, slope ∞ gives only $x = 0$.